

Cofinal strata and ordinal absorption

Finitary powersets of finite lexicographic sums

An exploratory research article

Prepared for Vladimir Reshetnikov with ChatGPT

19 September 2026

Abstract

We investigate a concrete instance of the published problem of enlarging classes of well-quasi-orders whose finitary powersets have compositionally computable ordinal invariants. For a finite poset Q , we stratify the finitely generated downsets of a lexicographic sum $\sum_{q \in Q}^{\text{lex}} P_q$ by the inclusion-maximal antichains of Q . Under explicit hypotheses on the fibers, each stratum has a pure maximal order type ω^{γ_A} . A cofinal-absorption theorem shows that its contribution survives exactly when no higher stratum has a strictly larger exponent.

For uniform ordinal fibers $\alpha = \omega^\rho$, $\rho > 0$, this yields

$$\mathfrak{o}(\mathcal{K}(S_\alpha(Q))) = \sum_{k=\mathfrak{w}(Q)}^1 \omega^{\rho \otimes k} c_k(Q), \quad \mathfrak{h}(\mathcal{K}(S_\alpha(Q))) = \alpha \cdot \mathfrak{h}(\mathcal{M}(Q)).$$

Here $c_k(Q)$ counts maximal antichains of size k not dominated by a larger one, and $\mathcal{M}(Q)$ is their domination poset. The first formula extends to nonuniform WPO fibers of prescribed pure finitary-powerset type. We also obtain a leading-coefficient counting reduction, an explicit maximal-chain construction, and a Cantor-normal-form truncation rule for finite nonuniform lexicographic sums. Complete proofs accompany exact-arithmetic Python code and finite structural checks. This is a partial answer to the broader research direction, not a solution of its general width or arbitrary-iteration problems; priority of the derived formulas is not independently established.

Scope of the claims. The mathematical results below are proved from stated standard WPO facts. The accompanying computations test finite identities and the implementation; they are not a formal verification of infinite ordinals. The general extension question remains broader than the class treated here.

Contents

1	The selected open direction and the precise target	1
2	Definitions and ordinal tools	2
3	Cofinal absorption	3
4	A finite cofinal-stratification theorem	4
5	The frontier poset of a finite order	5
6	Canonical strata of the finitary powerset	6
7	The general maximal-order-type formula	7
8	Uniform ordinal fibers: an ordinal polynomial	8
9	Exact height and an attaining chain	9
10	Examples and comparisons	10
11	Coefficients, counting and restricted reuse	12
12	Algorithm and reproducible verification	13
13	Boundaries and failed extensions	15
14	What remains open and what the attempt achieved	16
A	Finite nonuniform lexicographic sums	16
B	A compact proof-dependency audit	18

1 The selected open direction and the precise target

Abriola, Halfon, Lopez, Schmitz, Schnoebelen and Vialard ask, in the conclusion of *Measuring well-quasi-ordered finitary powersets*, which extensions of their elementary family retain computable ordinal invariants [1, Section 6, p. 23]. Their grammar uses disjoint unions, Cartesian products, words, multisets and finitary powersets. Arbitrary finite-poset lexicographic gluing is not one of its constructors. We take this missing constructor as a specific target.

Problem 1.1 (Finite-gluing instance). Let Q be a finite poset and let P_q , $q \in Q$, be WPOs. Determine the maximal order type and height of the finitary powerset of

$$S = \sum_{q \in Q}^{\text{lex}} P_q,$$

from finite information about Q and appropriate invariants of the fibers. In particular, obtain exact formulas when every fiber is the ordinal ω^ρ , for a fixed $\rho > 0$.

This formulation is our concrete subproblem, not a quotation of a separately numbered conjecture. The broad published question does not prescribe this particular class. The results here give an additional compositional rule with explicit hypotheses, rather than a claim that the whole elementary calculus has been closed under a new operation.

1.1 What is established here

The general maximal-order-type result is Theorem 7.3. Its hypotheses require that the fibers have no finite cofinal subsets and that $\text{o}(\mathcal{K}(P_q)) = \omega^{\beta_q}$, with $\beta_q > 0$. The exponents may vary with q , and the fibers need not be chains. The two uniform formulas are Corollary 8.1 and Theorem 9.1. The latter includes an actual chain attaining the computed height.

The main proof mechanism is independent of finitary powersets: a finite stratification with pure stratum types and upward cofinality admits an exact survivor formula. The finitary-powerset construction supplies a canonical such stratification. The distinction between a finite stratification and a lexicographic decomposition is essential: comparable strata need not have all their points comparable.

We do not determine ordinal width in general. Nor do we claim a formula for arbitrary repeated applications of $\mathcal{K}$, arbitrary impure fiber types, or infinite indexing posets. A concrete failure of a tempting nonuniform height formula is proved in Section 13.

1.2 Status and relation to existing results

The source for the open direction is the July 2024 version of [1]; its arXiv record and the final-page question were checked on 19 September 2026. A targeted literature search did not locate the frontier formulas proved below. That search is not a proof of novelty, and the results should be checked against specialist literature before priority is claimed.

A nearby problem must not be confused with our target: maximal order types of uniform lexicographic products have already been treated in Vialard's published 2025 paper [4]. We use that publication, rather than an older preprint's status statement, in assessing the literature. Appendix A derives a finite nonuniform formula as a supporting result and recovers the known finite uniform case. It is not presented as a resolution of an open lexicographic-product problem.

The background facts about maximal linearizations and natural operations are classical [2, 3]. The contribution attempted here is the cofinal-stratum reduction and its consequences for the chosen finite-gluing instance.

2 Definitions and ordinal tools

2.1 WPOs, rank and maximal order type

A *well-partial-order* (WPO) is a partial order in which every infinite sequence $x_0, x_1, \dots$ has $i < j$ with $x_i \leq x_j$. All posets considered below are sets, and we work in ordinary set theory with choice. A well-quasi-order can be replaced by its quotient under mutual comparison.

A *linearization* is a total order extending the partial order on the same underlying set. Every linearization of a WPO is a well-order. Its largest possible order type is denoted $o(P)$. The de Jongh–Parikh maximal-linearization theorem guarantees that this maximum is attained [2].

For a well-founded poset, define

$$r_P(x) = \sup_{y < x} (r_P(y) + 1), \quad h(P) = \sup_{x \in P} (r_P(x) + 1).$$

We use $h(\emptyset) = 0$. This is the height defined by the rank of the tree of finite strictly decreasing sequences. In particular, an ordinal chain α has height α . An order-preserving rank bound will mean a map into ordinals which is *strictly* increasing on strict comparisons; such a map bounds r_P pointwise by well-founded induction.

The ordinal width $w(P)$ is the rank of the tree of finite sequences of pairwise incomparable elements. When P has finite maximum antichain cardinality d , this ordinal width equals the integer d . We use width only in this finite sense for index posets and the input examples; no general output-width formula is asserted.

2.2 Finite downsets and the Hoare quotient

For a poset P , set

$$\mathcal{K}(P) = \{\downarrow F : F \subseteq P \text{ finite}\}, \quad \downarrow F = \{x \in P : \exists f \in F (x \leq f)\},$$

ordered by inclusion. The empty set belongs to $\mathcal{K}(P)$. Finite subsets have the Hoare relation

$$F \leq_H G \iff (\forall f \in F)(\exists g \in G) f \leq g.$$

This is equivalent to $\downarrow F \subseteq \downarrow G$, so $\mathcal{K}(P)$ is exactly its antisymmetric quotient. These are finitely generated *downsets*; they are not required to be directed ideals. Equivalently, they are the compact elements of the lattice of all downsets of P .

For a WPO P , $\mathcal{K}(P)$ is a WPO. One standard proof applies the finite-word WPO theorem to arbitrary enumerations of finite subsets: a word embedding implies Hoare domination. We write $\mathcal{K}(P)^+ = \mathcal{K}(P) \setminus \{\emptyset\}$.

A subset $C \subseteq P$ is *cofinal* if every $p \in P$ satisfies $p \leq c$ for some $c \in C$. Thus

$$P \text{ has no finite cofinal subset} \iff P \notin \mathcal{K}(P). \quad (1)$$

This condition ensures that a whole fiber cannot be generated internally by finitely many points.

2.3 Lexicographic sums

For pairwise disjoint nonempty posets P_q indexed by a finite poset Q , write $S = \sum_{q \in Q}^{\text{lex}} P_q$. Within a fiber we use its original order; for different fibers all points of P_q lie below all points of P_r exactly when $q < r$. Fibers over incomparable indices remain incomparable. The resulting poset is a WPO whenever every fiber is a WPO: an infinite sequence has an infinite subsequence in a single fiber.

For uniform ordinal fibers $P_q = \alpha$, denote this poset by $S_\alpha(Q)$. This notation avoids choosing between competing conventions for the order of factors in a lexicographic product.

2.4 Natural arithmetic and standard monotonicity

Unmarked $+$ and $\cdot$ are ordinary ordinal addition and multiplication. We use $\#$ for Hessenberg natural sum and $\otimes$ for natural product. In Cantor normal form, natural sum adds coefficients at equal exponents; natural product multiplies the corresponding formal polynomials, using natural sum for exponents. Consequently

$$\omega^\beta \otimes \omega^\gamma = \omega^{\beta \# \gamma}, \quad \underbrace{\rho \# \cdots \# \rho}_{k \text{ times}} = \rho \otimes k. \quad (2)$$

Natural sum is strictly increasing in each argument. For any β ,

$$\mu_1, \dots, \mu_t < \omega^\beta \implies \mu_1 \# \cdots \# \mu_t < \omega^\beta \quad (3)$$

when t is finite. For $\beta = 0$, all the μ_i are zero. For $\beta > 0$, this follows immediately from the finite Cantor normal forms: every appearing exponent is below β .

We use the following standard WPO facts [2, 3]:

$$o(P \sqcup R) = o(P) \# o(R), \quad o(P \times R) = o(P) \otimes o(R). \quad (4)$$

An induced subposet cannot have larger maximal order type. Adding comparisons on the same set cannot increase maximal order type, because it only removes possible linearizations. It follows that, for a finite partition $P = \sqcup_i P_i$ into induced subposets,

$$o(P) \leq \#_i o(P_i). \quad (5)$$

These background results are the nontrivial external inputs to our proofs. The new reductions below do not assume a general formula for the finitary-powerset operation.

3 Cofinal absorption

The first lemma explains why certain low-exponent strata can disappear entirely from a maximal order type.

Lemma 3.1 (Cofinal absorption). *Let X be a WPO with a disjoint partition $X = U \sqcup V$. Suppose that U is cofinal in X and*

$$o(U) = \kappa = \omega^\beta, \quad o(V) < \kappa.$$

Then $o(X) = \kappa$.

Proof. The lower bound follows because U is an induced subposet of X . For the upper bound, let L be any linearization of X . Suppose its order type is greater than κ . Let I be its initial segment consisting of the first κ positions.

The restriction of L to $U \cap I$ has order type at most κ . If it had type $\mu < \kappa$, the restriction to $V \cap I$ would have some type $\nu \leq o(V) < \kappa$. The order I is a linearization of the disjoint union of these two restricted well-orders. Hence

$$\kappa = \text{type}(I) \leq \mu \# \nu < \kappa,$$

by (4) and (3), a contradiction. Therefore $L \upharpoonright (U \cap I)$ has type exactly κ .

Choose x at a position after I . Cofinality gives $u \in U$ with $x \leq u$. Since L extends X , u also occurs after I . Thus $L \upharpoonright U$ contains its initial suborder of type κ and another point after it, so its type is at least $\kappa + 1$. This contradicts $o(U) = \kappa$. No linearization can exceed κ , proving the result. $\square$

Cofinality alone is insufficient when the cofinal part has an impure maximal type. For example, let U have type $\omega + 1$, let v be incomparable with the initial ω points of U , and put v below its last point. Then U is cofinal but $\text{o}(U \cup \{v\}) = \omega + 2$. The pure-power assumption in Lemma 3.1 is doing real work.

4 A finite cofinal-stratification theorem

Definition 4.1. A *finite cofinal stratification* of a WPO X consists of a finite poset I and a partition $X = \bigsqcup_{i \in I} X_i$ into nonempty induced subposets such that

$$x \in X_i, y \in X_j, x \leq y \implies i \leq j, \quad (6)$$

$$i \leq j \implies (\forall x \in X_i)(\exists y \in X_j) x \leq y. \quad (7)$$

The second condition is upward cofinality, not universal comparability.

Theorem 4.2 (Surviving strata). Suppose $X = \bigsqcup_{i \in I} X_i$ has a finite cofinal stratification and $\text{o}(X_i) = \omega^{\beta_i}$. Define

$$R = \{i \in I : \text{there is no } j > i \text{ with } \beta_j > \beta_i\}.$$

Then

$$\boxed{\text{o}(X) = \#_{i \in R} \omega^{\beta_i}.} \quad (8)$$

Moreover, some maximal linearization places each entire stratum in a single block.

Proof. Upper bound. For each $i \notin R$, choose $r(i) \geq i$ for which $\beta_{r(i)}$ is maximal among the exponents of successors of i , including i itself. Since i is not a survivor, this maximum is strictly larger than β_i . Furthermore, $r(i) \in R$: a successor of $r(i)$ with still larger exponent would contradict its choice.

For each $r \in R$, form the induced subposet

$$G_r = X_r \cup \bigcup_{\{i \notin R : r(i)=r\}} X_i.$$

These groups partition X . By upward cofinality, X_r is cofinal in G_r . By the partition bound and finite natural-sum closure,

$$\text{o}(G_r \setminus X_r) \leq \#_{\{i : r(i)=r\}} \omega^{\beta_i} < \omega^{\beta_r}.$$

The inequality also holds when the group has no other strata, since the left side is zero. Lemma 3.1 yields $\text{o}(G_r) = \omega^{\beta_r}$. Applying the partition bound once more proves $\text{o}(X) \leq \#_{r \in R} \omega^{\beta_r}$.

Lower bound. If $i, j \in R$ and $i < j$, then $\beta_i \geq \beta_j$; otherwise i would not survive. Order the elements of R by nonincreasing exponent, breaking ties by any linear extension of their induced order. This total order is compatible with the order inherited from I .

Any linear extension of an induced subposet of a finite poset extends to a linear extension of the whole poset. To see this directly, add the chosen comparisons between elements of the subposet. A directed cycle would give a chain of newly added comparisons, linked by old directed paths, whose endpoints lie in the subposet. Each old path already induces a comparison in its inherited order. Such a cycle would contradict compatibility of the chosen total order. The enlarged acyclic relation therefore has a linear extension.

Apply this observation to $R \subseteq I$. Concatenate maximal linearizations of all the X_i in the resulting order of I . Condition (6) ensures that this concatenation is a linearization of X . Its restriction to survivor blocks is an ordinary sum of the monomials ω^{β_i} in nonincreasing

exponent order. That ordinary sum equals their natural sum. The entire linearization has at least this order type, giving the lower bound. The upper bound already proved shows that the concatenation is maximal. $\square$

Remark 4.3. The theorem permits repeated exponents. A higher stratum of *equal* exponent does not absorb a lower one; both contributions survive unless a strictly larger exponent occurs still higher. This strict inequality is responsible for the finite coefficients in later formulas.

The proof also explains why an algorithm considering only block orders is not by itself enough. Such an algorithm supplies a lower bound. The cofinal-absorption argument controls all possible interleavings inside linearizations and supplies the matching upper bound.

5 The frontier poset of a finite order

Definition 5.1. For a nonempty finite poset Q , let $\mathcal{M}(Q)$ be its set of *inclusion-maximal* antichains. For $A \in \mathcal{M}(Q)$ put

$$D(A) = \{q \in Q : \exists a \in A (q < a)\}.$$

Order maximal antichains by domination:

$$A \leq B \iff (\forall a \in A)(\exists b \in B) a \leq b. \quad (9)$$

We call $(\mathcal{M}(Q), \leq)$ the *frontier poset*.

Maximality here is by inclusion, not cardinality. A maximal antichain can have fewer elements than the width of Q . This is exactly what happens in the two orientations of a three-point fork.

Lemma 5.2. For $A, B \in \mathcal{M}(Q)$,

$$A = \min(Q \setminus D(A)), \quad A \leq B \iff D(A) \subseteq D(B). \quad (10)$$

In particular, $A \mapsto D(A)$ is injective and $\leq$ is a partial order. The least and greatest frontiers are $\min Q$ and $\max Q$.

Proof. Every predecessor of an element of A lies in $D(A)$, while A itself is disjoint from $D(A)$. Conversely, suppose $x \in \min(Q \setminus D(A))$. By maximality, x is comparable with an element $a \in A$. It cannot lie strictly below a , since then $x \in D(A)$. It cannot lie strictly above a , since $a \notin D(A)$ would contradict minimality of x . Thus $x = a$.

If $A \leq B$, every strict predecessor of an $a \in A$ is a strict predecessor of some $b \in B$, proving $D(A) \subseteq D(B)$. Conversely, assume this inclusion and take $a \in A$. If $a \in D(B)$, it is strictly below some $b \in B$. Otherwise every strict predecessor of a lies in $D(A) \subseteq D(B)$, so $a \in \min(Q \setminus D(B)) = B$. Either way a is dominated by B .

Injectivity follows from the first identity. Every element of a finite poset lies above a minimal element and below a maximal one, giving the last assertion. $\square$

For the N -shaped poset with comparisons $a < c$, $b < c$, $b < d$, the frontiers form the three-element chain

$$\{a, b\} < \{a, d\} < \{c, d\}. \quad (11)$$

The middle frontier is not a level of the original height-two poset. This already suggests why its finitary powerset can have three ordinal height stages rather than two.

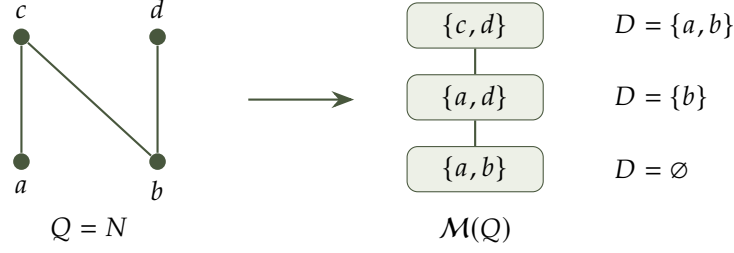


Figure 1: A height-two skeleton can have a height-three frontier poset. The right-hand labels are the completed-fiber sets.

6 Canonical strata of the finitary powerset

Let $S = \sum_{q \in Q}^{\text{lex}} P_q$, where Q is finite and nonempty, and assume that every nonempty fiber P_q has no finite cofinal subset. For $K \in \mathcal{K}(S)$ define its set of *completed fibers* by

$$D_K = \{q \in Q : P_q \subseteq K\}.$$

Lemma 6.1 (Canonical frontier). *Every $K \in \mathcal{K}(S)$ has a unique frontier $A \in \mathcal{M}(Q)$ with $D_K = D(A)$. For this frontier, K contains all of each fiber indexed by $D(A)$, intersects fibers indexed by A in finitely generated downsets, and is empty on all other fibers.*

Proof. Write $K = \downarrow F$ with F finite, let B be the set of indices meeting F , and let $A_0 = \max B$. For $F = \emptyset$, take $B = A_0 = \emptyset$.

If $q < a$ for some $a \in A_0$, a generator in P_a lies above every point of P_q , so $P_q \subseteq K$. If no generator has index strictly above q , then $K \cap P_q$ is generated internally by the finite set $F \cap P_q$. By (1), it cannot be all of P_q . Therefore

$$D_K = D(A_0) = \bigcup_{a \in A_0} \{q : q < a\}. \quad (12)$$

Write this downset as D and put $A = \min(Q \setminus D)$. Every member of A_0 belongs to A : its predecessors lie in D , and it does not lie in D itself. Conversely, every strict predecessor of an element of A lies in D . Thus $D(A) = D$, since $D = D(A_0) \subseteq D(A)$ as well.

The set A is a maximal antichain. An element outside D lies above a minimal element of $Q \setminus D$, while an element of D lies below some member of $A_0 \subseteq A$. Thus every element of Q is comparable with a member of A .

A nonempty intersection $K \cap P_q$ with $q \notin D$ requires a generator at index q itself, with no generator strictly above it. Hence $q \in A_0 \subseteq A$. These intersections are generated by $F \cap P_q$. The uniqueness of A follows from Lemma 5.2. $\square$

Let X_A be the set of all $K \in \mathcal{K}(S)$ with completed-fiber set $D(A)$. For $a \in A$, write $K_a = K \cap P_a$. The lemma gives a precise product representation, with one support condition:

$$X_A \cong \left\{ (K_a)_{a \in A} \in \prod_{a \in A} \mathcal{K}(P_a) : D(A) = \bigcup_{\{a: K_a \neq \emptyset\}} \{q : q < a\} \right\}. \quad (13)$$

The order is coordinatewise inclusion. To verify sufficiency, choose finite generators for the nonempty K_a and take their union inside S . The support condition makes its completed-fiber set exactly $D(A)$. Necessity follows from (12).

In particular, as induced subposets,

$$\prod_{a \in A} \mathcal{K}(P_a)^+ \subseteq X_A \subseteq \prod_{a \in A} \mathcal{K}(P_a). \quad (14)$$

Every X_A is nonempty, since all active coordinates can be chosen nonempty. The empty downset belongs to the stratum indexed by $\min Q$; there is no extra empty stratum.

Lemma 6.2 (Cofinal frontier stratification). *The partition*

$$\mathcal{K}(S) = \bigsqcup_{A \in \mathcal{M}(Q)} X_A$$

is a finite cofinal stratification indexed by $\mathcal{M}(Q)$.

Proof. If $K \subseteq L$, every fiber completed by K is completed by L . Thus $D_K \subseteq D_L$, and Lemma 5.2 gives the required comparison of frontier indices.

For upward cofinality, let $A \leq B$ and $K \in X_A$. Choose nonempty finitely generated downsets $L_b \in \mathcal{K}(P_b)^+$ for every $b \in B$. When $b \in A \cap B$, choose L_b to contain K_b : adjoin one point to its finite generating set if necessary. Use these coordinates and completed fibers $D(B)$ to form $L \in X_B$.

We have $D(A) \subseteq D(B)$. Moreover, each $a \in A \setminus B$ lies strictly below some $b \in B$, so its entire fiber is completed in L . The remaining active coordinates, indexed by $A \cap B$, were explicitly enlarged. Hence $K \subseteq L$. $\square$

The last argument chooses L in response to K . It does not say that *every* member of X_B dominates K . For example, for the skeleton consisting of an isolated point x and a chain $y < z$, the frontiers $\{x, y\} < \{x, z\}$ share an x -coordinate. A downset with a large x -coordinate in the first stratum is not below one with a small x -coordinate in the second. This is why replacing the strata by fully ordered blocks would give only a bound without Theorem 4.2.

7 The general maximal-order-type formula

Lemma 7.1 (Removing the least downset). *If $\text{o}(\mathcal{K}(P))$ is infinite, then $\text{o}(\mathcal{K}(P)^+) = \text{o}(\mathcal{K}(P))$.*

Proof. The empty downset is below everything else, so every linearization is a single least point followed by a linearization of $\mathcal{K}(P)^+$. Attainment of maximal linearizations gives $\text{o}(\mathcal{K}(P)) = 1 + \text{o}(\mathcal{K}(P)^+)$. The latter type must be infinite; for every infinite ordinal η , $1 + \eta = \eta$. $\square$

Remark 7.2 (Purity already excludes finite cofinality). In the next theorem the no-finite-cofinality assumption is stated to make the structural argument visible, but it follows from the positive pure type assumption. Indeed, a finite cofinal subset of P would make P itself the greatest member of $\mathcal{K}(P)$. Then

$$\text{o}(\mathcal{K}(P)) = \text{o}(\mathcal{K}(P) \setminus \{P\}) + 1$$

would be a successor ordinal, whereas ω^β is a limit for $\beta > 0$. Thus the numerical hypothesis alone implies this boundary condition (and also implies that P is nonempty).

Theorem 7.3 (Finite lexicographic gluing). *Let Q be a finite nonempty poset. For each $q \in Q$, let P_q be a nonempty WPO having no finite cofinal subset and satisfying*

$$o(\mathcal{K}(P_q)) = \omega^{\beta_q}, \quad \beta_q > 0.$$

For $A \in \mathcal{M}(Q)$ define

$$\gamma_A = \#_{a \in A} \beta_a, \quad \mathcal{R} = \{A \in \mathcal{M}(Q) : \nexists B > A (\gamma_B > \gamma_A)\}.$$

Then

$$o\left(\mathcal{K}\left(\sum_{q \in Q}^{\text{lex}} P_q\right)\right) = \#_{A \in \mathcal{R}} \omega^{\gamma_A}. \quad (15)$$

Proof. For each frontier A , the two outer products in (14) have the same maximal order type, by Lemma 7.1 and the Cartesian-product formula:

$$\bigotimes_{a \in A} \omega^{\beta_a} = \omega^{\#_{a \in A} \beta_a} = \omega^{\gamma_A}.$$

Induced-subposet monotonicity therefore gives $o(X_A) = \omega^{\gamma_A}$. Lemma 6.2 provides a finite cofinal stratification. Apply Theorem 4.2. $\square$

The theorem is compositional in a precise, limited sense. Once the pure values $o(\mathcal{K}(P_q))$ are known and the no-finite-cofinality hypotheses are verified, no additional internal data about the fibers enter the output maximal order type. All remaining work takes place in the finite weighted frontier poset. This does not supply a method for discovering the required fiber values for arbitrary input WPOs.

Example 7.4 (A nonuniform weighted frontier). Take the N -poset from (11) and put

$$(\beta_a, \beta_b, \beta_c, \beta_d) = (3, 1, 1, 2).$$

The successive frontier exponents are 4, 5, 3. The first is absorbed by the second, while the second and third survive. Therefore

$$o(\mathcal{K}(S)) = \omega^5 + \omega^3.$$

This is realized, for example, by the ordinal fibers $(\omega^3, \omega, \omega, \omega^2)$.

8 Uniform ordinal fibers: an ordinal polynomial

Fix $\rho > 0$ and $\alpha = \omega^\rho$. Finitely generated downsets of the chain α are the empty set and principal downsets $\downarrow \xi$ for $\xi < \alpha$. Thus

$$\mathcal{K}(\alpha) \cong 1 + \alpha \cong \alpha. \quad (16)$$

The ordinal α is a nonzero limit and has no finite cofinal subset. All hypotheses of Theorem 7.3 hold with $\beta_q = \rho$.

Define the nonnegative integers

$$c_k(Q) = \#\{A \in \mathcal{M}(Q) : |A| = k \text{ and } (\forall B \geq A) |B| \leq k\}. \quad (17)$$

The notation $\#$ outside ordinal operations in this display denotes ordinary finite cardinality. In words, $c_k(Q)$ counts size- k maximal antichains not dominated by a larger maximal antichain.

Corollary 8.1 (Uniform maximal order type). *For a finite nonempty poset Q of width W and $\alpha = \omega^\rho$, $\rho > 0$,*

$$\mathfrak{o}(\mathcal{K}(S_\alpha(Q))) = \omega^{\rho \otimes W} \cdot c_W(Q) + \omega^{\rho \otimes (W-1)} \cdot c_{W-1}(Q) + \cdots + \omega^\rho \cdot c_1(Q). \quad (18)$$

Terms with zero coefficients are omitted. The displayed additions and coefficient multiplications are ordinary ordinal operations.

Proof. The exponent of a frontier is $\rho \otimes |A|$, by (2). Since $\rho > 0$, this is strictly increasing with the positive integer $|A|$. Therefore a frontier survives precisely under the condition in (17). Collect equal exponents in Theorem 7.3 and list them in decreasing order. $\square$

For $\rho = 1$, the output is literally a finite Cantor-normal-form polynomial in ω whose coefficients depend on the finite skeleton:

$$\mathfrak{o}(\mathcal{K}(S_\omega(Q))) = \sum_{k=W}^1 \omega^k c_k(Q).$$

Thus the infinite ordinal calculation separates into a finite combinatorial statistic and an explicit substitution in ordinal arithmetic.

Corollary 8.2 (Leading coefficient). *The coefficient $c_W(Q)$ is the number of antichains of Q having maximum cardinality W .*

Proof. Every maximum-cardinality antichain is inclusion-maximal, and none can be dominated by an antichain of larger cardinality. Conversely, every frontier counted by c_W has cardinality W . $\square$

Remark 8.3 (Natural multiplication cannot be replaced). For $\rho = \omega + 1$ and a two-point antichain Q ,

$$\rho \otimes 2 = \omega \cdot 2 + 2, \quad \rho \cdot 2 = \omega \cdot 2 + 1.$$

Consequently the maximal order type is $\omega^{\omega \cdot 2 + 2}$, not $\omega^{\omega \cdot 2 + 1}$. The code keeps these two multiplication operations separate.

9 Exact height and an attaining chain

Let

$$\ell(Q) = h(\mathcal{M}(Q)),$$

where height of the finite frontier poset means the largest number of elements in one of its chains.

Theorem 9.1 (Uniform height). *If Q is finite and nonempty and $\alpha = \omega^\rho$ with $\rho > 0$, then*

$$\mathfrak{h}(\mathcal{K}(S_\alpha(Q))) = \alpha \cdot \ell(Q). \quad (19)$$

There is a chain of this exact order type inside $\mathcal{K}(S_\alpha(Q))$.

Proof. Upper bound. Choose an order isomorphism $\theta : \mathcal{K}(\alpha) \rightarrow \alpha$ from (16). For a frontier A , let $d(A)$ be the largest number of strict steps in a chain ending at A in $\mathcal{M}(Q)$. Then $0 \leq d(A) < \ell(Q)$, and $A < B$ implies $d(A) < d(B)$.

For $K \in X_A$, define

$$t_A(K) = \#_{a \in A} \theta(K_a), \quad R(K) = \alpha \cdot d(A) + t_A(K).$$

All coordinates $\theta(K_a)$ are below α , so finite natural-sum closure gives $t_A(K) < \alpha$. Thus $R(K) < \alpha \cdot \ell(Q)$.

If $K \subsetneq L$ lie in the same stratum, their active coordinate tuples are strictly ordered componentwise. Strict monotonicity of natural sum gives $t_A(K) < t_A(L)$. If their frontiers are distinct, say $A < B$, then

$$R(K) < \alpha \cdot (d(A) + 1) \leq \alpha \cdot d(B) \leq R(L).$$

The index-respecting property rules out all other cases. Hence R is strictly increasing on strict inclusion, so well-founded induction yields $r_{\mathcal{K}(S)}(K) \leq R(K)$. Since $R(K) < \alpha \cdot \ell(Q)$ for every K , the definition of height gives the required upper bound.

Lower bound and attainment. Choose a longest frontier chain

$$A_0 < A_1 < \cdots < A_{\ell-1}, \quad \ell = \ell(Q).$$

For each $i < \ell - 1$, choose $a_i \in A_i \setminus A_{i+1}$. Such a point exists: if one inclusion-maximal antichain were properly contained in another, it would not be maximal. Since $A_i \leq A_{i+1}$ and $a_i \notin A_{i+1}$, we have $a_i \in D(A_{i+1})$. It belongs to every later completed-fiber set as well. At the last stage choose any $a_{\ell-1} \in A_{\ell-1}$.

For each i and $\xi < \alpha$, form $K_{i,\xi} \in X_{A_i}$ by completing all fibers in $D(A_i)$, setting every active coordinate other than a_i to $\downarrow 0$, and setting the a_i -coordinate to $\downarrow \xi$. Every active coordinate is nonempty, so these are valid members of the stratum. For fixed i , the family $(K_{i,\xi})_{\xi < \alpha}$ is a chain of type α .

If $i < j$, the varying coordinate a_i is completed at stage j . Any other coordinate in A_i is either completed at stage j or remains an active coordinate there, whose downset contains $\downarrow 0$. Already completed fibers stay completed. It follows that $K_{i,\xi} \subseteq K_{j,\eta}$ for all $\xi, \eta < \alpha$. Inclusion is strict because $D(A_i) \subsetneq D(A_j)$: some fiber is fully present at stage j but not at stage i .

The ordered concatenation of these ℓ many chains is therefore a chain of type $\alpha \cdot \ell$. This proves the lower bound and attainment. $\square$

The height and maximal-order-type formulas inspect different aspects of the same finite object. Height records the longest chain of frontiers. Maximal order type records the exponents of frontiers that cannot be absorbed from above. Neither statistic determines the other.

10 Examples and comparisons

10.1 Chains, antichains and oriented forks

If Q is a chain of n elements, its frontiers are its n singleton subsets, ordered in a chain. Every frontier survives, so both output invariants are $\alpha \cdot n$. This agrees with $S_\alpha(Q) \cong \alpha \cdot n$ and $\mathcal{K}(\alpha \cdot n) \cong \alpha \cdot n$.

If Q is an antichain of n elements, it has only one maximal antichain, namely Q itself. The output has maximal order type $\omega^{\rho \otimes n}$ and height α . Directly, it is the Cartesian product of n copies of $\mathcal{K}(\alpha) \cong \alpha$.

For the upward fork $a < b, c$, the frontiers are $\{a\} < \{b, c\}$. The size-one frontier is absorbed, giving

$$o(\mathcal{K}(S_\omega(Q))) = \omega^2, \quad h(\mathcal{K}(S_\omega(Q))) = \omega \cdot 2.$$

For the downward fork $a, b < c$, the frontiers are $\{a, b\} < \{c\}$. Both survive, giving

$$o(\mathcal{K}(S_\omega(Q))) = \omega^2 + \omega, \quad h(\mathcal{K}(S_\omega(Q))) = \omega \cdot 2.$$

Reversing a finite skeleton need not preserve the output maximal order type, even though it preserves its size, ordinary height and width.

10.2 Three indistinguishable inputs with different outputs

Consider the following four-point height-two skeletons, all with lower points a, b and upper points c, d :

$$\begin{array}{l|l} Q_2 & a < c, a < d, b < c, b < d \\ Q_3 & a < c, b < c, b < d \\ Q_4 & a < c, b < d. \end{array}$$

The middle one is the N -poset. Their frontier posets have respectively two, three and four elements; all frontiers have size two. The third frontier poset is a square rather than a four-element chain.

Skeleton	$ \mathcal{M}(Q) $	$\ell(Q)$	$\mathbf{o}(\mathcal{K}(S_\omega(Q)))$	$\mathbf{h}(\mathcal{K}(S_\omega(Q)))$
Complete two levels Q_2	2	2	$\omega^2 \cdot 2$	$\omega \cdot 2$
N -poset Q_3	3	3	$\omega^2 \cdot 3$	$\omega \cdot 3$
Two disjoint chains Q_4	4	3	$\omega^2 \cdot 4$	$\omega \cdot 3$

Table 1: Equal input invariants do not force equal finitary-powerset invariants. These values are consequences of the proved formulas.

For every one of these inputs,

$$(\mathbf{o}(S_\omega(Q_i)), \mathbf{h}(S_\omega(Q_i)), \mathbf{w}(S_\omega(Q_i))) = (\omega \cdot 4, \omega \cdot 2, 2). \quad (20)$$

Indeed, disjoint-union monotonicity bounds the input maximal type by $\omega \cdot 4$, and concatenating the four fibers in a linear extension attains it. The height is $\omega \cdot 2$, by a two-level rank bound and any chain of two comparable fibers. An antichain has at most one point in each fiber and its indices must be an antichain of Q_i , giving width two.

The general failure of functionality of finitary-powerset invariants was already proved in [1]; we do not claim that observation as new. Table 1 supplies a transparent finite-skeleton instance and explains the difference by explicit frontier statistics.

10.3 Disjoint unions of finite chains

Suppose Q is the disjoint union of k nonempty finite chains of lengths $n_1, \dots, n_k$. A maximal antichain chooses exactly one point from each chain. Hence

$$\mathcal{M}(Q) \cong \prod_{i=1}^k n_i, \quad |\mathcal{M}(Q)| = \prod_{i=1}^k n_i, \quad \ell(Q) = 1 + \sum_{i=1}^k (n_i - 1).$$

All frontiers have size k and survive. Thus

$$\begin{aligned} \mathbf{o}(\mathcal{K}(S_\alpha(Q))) &= \omega^{\rho \otimes k} \cdot \prod_{i=1}^k n_i, \\ \mathbf{h}(\mathcal{K}(S_\alpha(Q))) &= \alpha \cdot \left(1 + \sum_{i=1}^k (n_i - 1) \right). \end{aligned}$$

The maximal-type result also follows directly from $\mathcal{K}(S_\alpha(Q)) \cong \prod_i (\alpha \cdot n_i)$ and natural multiplication. The frontier derivation additionally explains the height coefficient.

11 Coefficients, counting and restricted reuse

11.1 Counting finite downsets in a leading coefficient

Proposition 11.1 (An explicit counting reduction). *Let R be a nonempty finite poset with n elements. Construct a height-two poset Q_R on $\{x^- : x \in R\} \cup \{x^+ : x \in R\}$ by*

$$x^- < y^+ \iff x \leq_R y,$$

and no other strict comparisons. Then $w(Q_R) = n$, and the coefficient of ω^n in $o(\mathcal{K}(S_\omega(Q_R)))$ is the number of order ideals of R , where an order ideal here means a downward-closed subset.

Proof. The n comparisons $x^- < x^+$ partition Q_R into n two-element chains, so every antichain has size at most n . All minus points form an antichain of size n , proving the width assertion.

An antichain of size n must choose exactly one of x^- , x^+ for each $x \in R$. Let I be the set of x for which it chooses x^+ . The chosen set is an antichain precisely when no $x \notin I$, $y \in I$ satisfy $x \leq_R y$. This is precisely the downward-closure condition on I . The construction is reversible: from an order ideal I , choose x^+ for $x \in I$ and x^- otherwise. Hence maximum antichains of Q_R are in bijection with order ideals of R . Apply Corollary 8.2. $\square$

This does not assert a computational-complexity classification without an additional counting-complexity theorem. It does show directly that an exact leading coefficient can encode a substantial finite counting problem. Even when the transfinite part of the answer is a single exponent, the coefficient is not generally a simple function of size, height and width of the skeleton.

11.2 When the output remains a single pure power

Proposition 11.2 (Pure-output criterion). *Under the hypotheses of Theorem 7.3, let $T = \max Q$, the greatest frontier. The output maximal order type is a single pure power of ω , with coefficient one, if and only if*

$$\gamma_T > \gamma_A \quad \text{for every } A \neq T. \quad (21)$$

In that case it equals ω^{γ_T} . For uniform fibers, (21) says that $\max Q$ is the unique maximum-cardinality antichain.

Proof. If (21) holds, T absorbs every other frontier and is the only survivor. Conversely, suppose some $A \neq T$ has $\gamma_A \geq \gamma_T$. If A is not already a survivor, repeatedly move to a strict successor of larger exponent. Finiteness forces this process to end at a survivor B . Its exponent is at least γ_A , and $B \neq T$: if a strict increase occurred its exponent exceeds γ_T , while if none occurred $B = A \neq T$. Since T always survives, there are at least two nonzero monomial contributions. Their natural sum cannot be a single pure power with coefficient one. The uniform assertion follows from strict increase of $\rho \otimes k$ in k . $\square$

If a finite lexicographic sum S satisfies this criterion, it can itself be used as a fiber in another application of Theorem 7.3. Indeed, $o(\mathcal{K}(S))$ has the required pure form. Also S has no finite cofinal subset: restrict any proposed finite cofinal set to a maximal index fiber; points outside that fiber cannot dominate its elements, contradicting the hypothesis there.

This is a genuine restricted reuse rule for the gluing construction. It must not be confused with unrestricted iteration of the powerset constructor: using $\mathcal{K}(S)$ as a new fiber would require information about $o(\mathcal{K}(\mathcal{K}(S)))$, which has not been supplied by the criterion.

12 Algorithm and reproducible verification

12.1 Exact finite procedure

For $n = |Q|$ and $m = |\mathcal{M}(Q)|$, a direct algorithm first computes transitive closure and enumerates maximal antichains. It then computes $D(A)$, builds the inclusion order on these downsets, and assigns each frontier its exponent γ_A . A frontier is discarded exactly when a strict successor has a larger exponent. Equal exponents of retained frontiers are collected in Cantor normal form.

Listing 1: The survivor calculation; ordinary and natural operations stay distinct.

```
frontiers = inclusion_maximal_antichains(Q)
D = {A: strict_predecessors_of(A) for A in frontiers}
gamma = {A: natural_sum(beta[a] for a in A) for A in frontiers}
retained = [A for A in frontiers
             if not any(D[A] < D[B] and gamma[A] < gamma[B]
                        for B in frontiers)]
answer = natural_sum(omega_power(gamma[A]) for A in retained)
```

The routine for uniform fibers additionally computes the longest-chain length in the finite frontier poset. This produces the height coefficient and a standard predecessor-pointer implementation can recover a longest frontier chain for the explicit construction in Theorem 9.1.

Naive enumeration costs $O(n^2 2^n)$ elementary relation operations. Building and comparing all frontier downsets costs $O(m^2 n)$ without bit-packing; transitive closure costs $O(n^3)$. Ordinal arithmetic and comparison have their own cost in the chosen notation representation. With exponents regarded as supplied exact data, a convenient bound on the finite combinatorial part is

$$O(n^3 + n^2 2^n + m^2 n).$$

No polynomial-time claim in n is made. The simple implementation is intended as an auditable reference, not an optimized maximal-antichain enumerator.

The mathematical theorems allow arbitrary ordinal exponents. The supplied program represents hereditary Cantor normal forms below ε_0 , using integers and recursively represented exponents, with no floating-point arithmetic. This restriction is an implementation boundary, not a theorem hypothesis. Ordinals at or above ε_0 require a richer notation system or abstract ordinal operations.

12.2 Independent block-order oracle

A second routine computes the greatest ordinary sum obtainable by ordering pure blocks in a linear extension of a finite index poset. For a subset J of indices it uses the recurrence

$$B(\emptyset) = 0, \quad B(J) = \max_{i \in \max J} (B(J \setminus \{i\}) + \omega^{\beta_i}). \quad (22)$$

The last block must be maximal in the remaining index order, so the recurrence exhausts all block linearizations. It is independent of the survivor rule, though not independent of ordinary ordinal arithmetic. Equality of the two computations checks the finite optimization identity. Only the absorption proof upgrades it to a statement about all linearizations of the infinite WPO.

12.3 What was actually checked

The delivered verification run used Python 3.13.5 and passed all 216,386 assertions. The machine-readable report is `data/verification.json`. It records the following coverage.

Check family	Cases
Naturally labelled posets on one through six vertices	5,231
Nonuniform frontier weights from $\{1, 2, 3\}$, through four vertices	3,450
General pure-stratum exponents from $\{0, 1, 2\}$, through four vertices	3,450
Hereditary-CNF weighted examples	24
Canonical finite profiles, over 50 skeletons	1,222
Pairs of profiles checked against Hoare domination	39,370
Deterministic series-parallel constructions	500
Order-ideal / maximum-antichain counting reductions	407
Nonuniform lexicographic CNF cases against expanded-block DP	3,450
Random exact ordinal-arithmetic triples, with fixed seed	500

Table 2: Coverage of the finite regression suite. These are not counts of isomorphism classes or formal-proof obligations.

The naturally labelled census has counts 1, 2, 7, 40, 357, 4824 for sizes one through six. A relation is naturally labelled when $i <_Q j$ implies $i < j$ as integers. Every finite poset admits such a labelling, but isomorphic copies can occur more than once. The census is therefore complete for this labelled convention, not an isomorphism-reduced enumeration.

The profile checks independently generate downsets from subsets of two sample generator positions per fiber, then compare them with the canonical stratum description. A distinguished symbol denotes an *entire* completed ω -fiber; it is not identified with the largest finite sample value. These checks verify the support condition, the equivalence of Hoare domination and profile inclusion, index monotonicity, and the constructed upward-cofinal witnesses on those samples.

The series-parallel tests compare the frontier formula with recursive ordinary addition for ordered sums and natural multiplication for disjoint unions. The final test family checks Appendix A by expanding finite Cantor normal forms into pure blocks and applying (22). Exact test inputs, fixed seeds and the full finite census are included in the archive.

12.4 Reproduction and file map

The project has no third-party Python dependencies. From its root run

```
python code/verify.py
python code/frontiers.py data/N.json
latexmk -pdf article.tex
```

The default verifier regenerates its JSON report and CSV census. Running `frontiers.py` without an input file prints the named examples. The JSON format uses integers for finite ordinals and lists of `[exponent, coefficient]` pairs for a nonfinite Cantor normal form. For example, $\omega + 1$ is represented by `[[1, 1], [0, 1]]`.

`code/ordinals.py` implements exact arithmetic; `code/frontiers.py` contains finite-poset and invariant routines; `code/verify.py` contains the verification suite. `PROOF_AUDIT.md` lists proof dependencies and boundaries, while `SOURCES.md` records the literature-status checks.

13 Boundaries and failed extensions

13.1 Why uniform height is a real restriction

It is tempting to extend the height theorem by assigning each frontier its stratum height and taking the greatest sum along a frontier chain. That fails, even with three ordinal fibers.

Let Q consist of an isolated point x and a chain $y < z$, with

$$P_x = \omega^2, \quad P_y = P_z = \omega.$$

There are two frontiers $\{x, y\} < \{x, z\}$, and each stratum has height ω^2 . The naive chain-sum rule predicts $\omega^2 \cdot 2$. But the whole input is the disjoint union $\omega^2 \sqcup (\omega \cdot 2)$, so

$$\mathcal{K}(S) \cong \omega^2 \times (\omega \cdot 2).$$

For (ξ, η) in this product, $\xi \# \eta < \omega^2$. This natural sum is strictly increasing under strict componentwise comparison, giving height at most ω^2 . A copy of ω^2 in the first coordinate gives the reverse inequality. Thus

$$h(\mathcal{K}(S)) = \omega^2 \neq \omega^2 \cdot 2. \quad (23)$$

The dominant x -coordinate persists across the frontier change and cannot be reset. In the uniform proof, one could always choose a *disappearing* coordinate with full type α at every transition. Here the disappearing coordinate has only type ω , not ω^2 . This pinpoints the obstruction rather than merely exhibiting a numerical failure. The general maximal-type theorem remains valid and gives $\omega^3 \cdot 2$, agreeing with natural multiplication.

13.2 Finite cofinality and impure types

If a fiber has a finite cofinal subset, it may become fully present without any generator in a strict successor fiber. Equation (12) then fails. For the one-point skeleton with a one-point fiber, $\mathcal{K}(S)$ has two elements, and one of its downsets completes the sole fiber. The only maximal antichain has empty strict downset, so completed fibers no longer index the proposed strata. This example violates the theorem's hypotheses; it is a warning against silently dropping them.

Likewise, replacing pure values ω^{β_q} by arbitrary ordinal values inside the same survivor formula is unjustified. The elementary counterexample after Lemma 3.1 shows that small terms may survive beneath the terminal part of an impure cofinal component. A successful extension will need to track relevant cuts of the components, not merely their leading exponents.

13.3 Finite indexing is also essential to the argument

Finiteness is used in three separate places: maximal antichains provide a finite stratification; absorbed strata can be grouped into finitely many lower-exponent pieces; and survivor block orders extend by a finite ordering argument. For infinitely many lower-exponent pieces, their natural-sum bound need not stay below a given ω^β . The present proof therefore does not justify replacing the finite index poset by an arbitrary WPO, even when individual frontiers are finite.

The empty index poset is harmless but needs a separate convention: $S = \emptyset$, so $\mathcal{K}(S) = \{\emptyset\}$ and both output invariants are 1. The main formulas explicitly assume $Q \neq \emptyset$; the implementation handles the empty case separately.

14 What remains open and what the attempt achieved

Problem 1.1 is answered for maximal order type under the pure-type hypotheses, and for height with uniform additively indecomposable ordinal fibers. The finite-skeleton information required is explicit: the domination order on inclusion-maximal antichains, their weighted exponents, and its longest-chain length.

A natural next mathematical target is a nonuniform height formula that tracks which coordinates disappear on each frontier transition. The counterexample in (23) shows that a scalar stratum height is not sufficient for a naive chain-sum recurrence. The canonical product profiles retain enough information to formulate more refined rank bounds, but no exact general formula is claimed here.

Ordinal width remains a separate target. The inequalities $w(X) \leq o(X)$ and the known invariant theory provide bounds, but the cofinal-absorption lemma does not automatically apply to antichain-tree rank: adding cross-stratum comparisons destroys antichains in ways not captured solely by successor exponents. Proving a corresponding width formula requires another argument.

A third direction is to replace the pure-type assumption by a finite collection of component cuts with controlled cofinality. The successful finite-CNF argument for actual lexicographic sums in Appendix A suggests what information may matter, but arbitrary WPO strata need not inherit the same cut structure from a chosen maximal linearization.

The present outcome is an exact, proof-backed partial answer to the published extension direction. Its main point is that a transfinite optimization problem reduces to a finite weighted frontier problem, with a separate cofinality proof ruling out larger interleavings. This claim is narrower than solving all finitary-powerset invariants, and stronger than an experimentally guessed formula for a few examples.

A Finite nonuniform lexicographic sums

This appendix gives a useful consequence of the same absorption theorem. It concerns the lexicographic sum itself, not its finitary powerset.

For a nonzero ordinal

$$\eta = \omega^{\delta_1} a_1 + \cdots + \omega^{\delta_s} a_s, \quad \delta_1 > \cdots > \delta_s,$$

define $\deg(\eta) = \delta_1$ and

$$\mathrm{tr}_{\geq \tau}(\eta) = \sum_{\{j: \delta_j \geq \tau\}} \omega^{\delta_j} a_j.$$

The latter is the initial Cantor-normal-form part whose exponents are at least τ .

Lemma A.1 (Replacing a fiber by a maximal linearization). *For a finite poset Q and WPOs P_q , let $\eta_q = o(P_q)$. Then*

$$o\left(\sum_{q \in Q}^{\mathrm{lex}} P_q\right) = o\left(\sum_{q \in Q}^{\mathrm{lex}} \eta_q\right).$$

Empty fibers can be deleted from the index poset, retaining its induced order on the remaining indices.

Proof. Choose maximal linearizations of the fibers. Replacing their orders by these linearizations adds comparisons to the lexicographic sum, so the right-hand side is at most the left-hand side.

Conversely, take an arbitrary linearization L of the original sum. Its restriction to P_q has some type $\zeta_q \leq \eta_q$. After identifying each restricted fiber with its ordinal type, L is a linearization of $\sum_q^{\text{lex}} \zeta_q$. This sum embeds as an induced subposet in $\sum_q^{\text{lex}} \eta_q$, by initial-segment embeddings within each fiber. Consequently

$$\text{type}(L) \leq o\left(\sum_q^{\text{lex}} \zeta_q\right) \leq o\left(\sum_q^{\text{lex}} \eta_q\right).$$

Taking the maximum over L proves the reverse inequality. $\square$

Theorem A.2 (Successor-degree truncation). *Let Q be finite and let P_q be WPOs with $\eta_q = o(P_q)$. Ignore indices whose fibers are empty. For every remaining q , put*

$$\tau(q) = \max(\{0\} \cup \{\deg(\eta_r) : r > q, \eta_r > 0\}).$$

Then

$$o\left(\sum_{q \in Q}^{\text{lex}} P_q\right) = \#_{\{q: \eta_q > 0\}} \text{tr}_{\geq \tau(q)}(\eta_q). \quad (24)$$

Proof. By Lemma A.1, it suffices to use the ordinal fibers η_q . Expand each of their finite Cantor normal forms into a finite sequence of pure blocks, repeating a block according to its coefficient. Within a fiber these blocks appear in nonincreasing exponent order. Between fibers, order every block of q below every block of r whenever $q < r$. This produces a finite pure-block stratification with full comparisons between comparable blocks, hence in particular a finite cofinal stratification.

A block of exponent δ is not absorbed by a later block of its own fiber, since internal exponents never increase. It is absorbed by a block of a strict successor fiber exactly when some such fiber has leading exponent greater than δ . Therefore it survives exactly when $\delta \geq \tau(q)$. Theorem 4.2 now says to take the natural sum of all surviving blocks, which is (24). $\square$

A.1 Recovery of the finite uniform formula

Suppose all n fibers have the same nonzero type $\eta = \omega^\delta c + \sigma$, where $\sigma < \omega^\delta$, and let k be the number of maximal elements of Q . Every nonmaximal fiber loses its tail σ , while every maximal fiber retains it. Thus

$$o\left(\sum_{q \in Q}^{\text{lex}} P\right) = \omega^\delta \cdot (cn) + (\sigma \otimes k).$$

This is the finite-index specialization of the uniform lexicographic product result treated in [4]. For finite η , the same expression simply says ηn . Its appearance here is a consistency check and a consequence of the more general finite-block argument, not an assertion that the uniform problem remained open.

A.2 A small tail example

For two incomparable fibers of types $\omega + 1$ and $\omega + 2$, no term is truncated, giving $\omega \cdot 2 + 3$. If the first fiber is below the second, its constant term is deleted, giving $\omega \cdot 2 + 2$. This is why ordinary concatenation of whole impure fibers is not a sufficient lower-bound construction for arbitrary partial index orders: maximal linearizations may separate their Cantor-normal-form blocks. The pure-stratum theorem avoids that difficulty by starting at the appropriate block granularity.

B A compact proof-dependency audit

The proof graph has three standard inputs: maximal-linearization attainment, the disjoint-union/natural-sum formula, and the Cartesian-product/natural-product formula. Induced-subposet monotonicity and elementary Cantor-normal-form arithmetic are used throughout.

From these, Lemma 3.1 proves the crucial upper bound for a cofinal pure component. Theorem 4.2 adds finite grouping and an explicit block-order lower bound. Lemmas 5.2, 6.1 and 6.2 establish the finite frontier model without any ordinal estimate. The product sandwich then feeds Theorem 4.2, yielding Theorem 7.3.

The height proof is separate. Its upper bound is an explicit ordinal rank map, and its lower bound is an explicit chain. It does not infer height from maximal order type or from a finite numerical test. The counting and pure-output results are finite corollaries of the frontier formula. Finally, Appendix A uses maximal linearizations only after proving that fiber replacement preserves the quantity being calculated.

No step depends on a claim that cofinality preserves arbitrary maximal order types, that frontier strata are fully comparable, that finite cutoffs compute infinite ranks, or that all ordinal widths equal maximal order types. The counterexamples and limitations in Section 13 are included to make these potential failure modes explicit.

References

- [1] Sergio Abriola, Simon Halfon, Aliaume Lopez, Sylvain Schmitz, Philippe Schnoebelen and Isa Vialard. *Measuring well-quasi-ordered finitary powersets*. arXiv:2312.14587v2, 17 July 2024. Open extension question: Section 6, p. 23. <https://arxiv.org/abs/2312.14587>.
- [2] D. H. J. de Jongh and Rohit Parikh. Well-partial orderings and hierarchies. *Indagationes Mathematicae*, 39(3):195–207, 1977.
- [3] Mirna Džamonja, Sylvain Schmitz and Philippe Schnoebelen. On ordinal invariants in well quasi orders and finite antichain orders. In *Well Quasi-Orders in Computation, Logic, Language and Reasoning*, Trends in Logic 53, pp. 29–54. Springer, 2020. Updated preprint: arXiv:1711.00428v5, 28 January 2025. <https://arxiv.org/abs/1711.00428>.
- [4] Isa Vialard. On maximal order type of the lexicographic product. *Logic Journal of the IGPL*, 33(4):jzaf051, 2025. Published 17 July 2025. <https://doi.org/10.1093/jigpal/jzaf051>.